

For some of the questions below, you may want to check the great slides by Torsten Hahmann (CSC 384 at University of Toronto) on **Skolemization, Most General Unifiers, First-Order Resolution**. It includes the Skolemization process to remove existential quantification, the algorithm for finding MGUs, and FOL resolution; all with great fully detailed examples!

Predicates

A *term* is either a variable or a name (sometimes called a constant).¹

A *formula* (or first-order formula) is

- $p(x_1, \dots, x_n)$ where $x_1, \dots, x_n$ are terms. p is called a *predicate*, and $p(x_1, \dots, x_n)$ is called an *atom*.
- $\neg P$ where P is a formula.
- $P \vee Q$, $P \wedge Q$, $P \Rightarrow Q$ and $P \Leftrightarrow Q$ where P and Q are formulae.
- $\forall x P$ and $\exists x P$ where P is a formula with free variable x . $\forall$ and $\exists$ are called *quantifiers*.

A *literal* is an atom or the negation of an atom. For example, $p(1, 2)$ and $\neg q(3)$ are both literals, but $\forall x \neg p(x)$ is not. We will say a formula is in *literal form* if all negations in the formula occur only in literals. For example, the formula $\neg \forall x p(x, x)$ is not in literal form, whereas $\neg p(1, 2)$ is in literal form.

Note that all the previous logical equivalences (denoted $\equiv$) are still true. The ones from last week and some specific to quantifiers are below.

$P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R)$	<i>Distributive Law 1</i>
$P \vee (Q \wedge R) \equiv (P \vee Q) \wedge (P \vee R)$	<i>Distributive Law 2</i>
$\neg \neg P \equiv P$	
$\neg(P \wedge Q) \equiv (\neg P) \vee (\neg Q)$	<i>De Morgan Law 1</i>
$\neg(P \vee Q) \equiv (\neg P) \wedge (\neg Q)$	<i>De Morgan Law 2</i>
$\forall x p(x) \equiv \forall y p(y)$	$\exists x p(x) \equiv \exists y p(y)$
$\forall x \forall y P \equiv \forall y \forall x P$	$\exists x \exists y P \equiv \exists y \exists x P$
$\neg \forall x P \equiv \exists x \neg P$	$\neg \exists x P \equiv \forall x \neg P$
$\forall x P(x) \Rightarrow Q(x) \equiv \forall x \neg Q(x) \Rightarrow \neg P(x)$	<i>Contrapositive</i>

Please remember that $\equiv$ is NOT a logical symbol, so $\alpha \equiv \beta$ is NOT a formula in the FOL language. Instead, $\alpha \equiv \beta$ is a shorthand for the claim (at the English meta-level), that $\alpha \models \beta$ and $\beta \models \alpha$, or what is the same, that for every FOL interpretation/models M , $M \models \alpha$ if and only if $M \models \beta$.

¹Strictly speaking there are other kinds of terms as well, but we will not be concerned with them in this course.

Using quantifiers and connectives

The choice of quantifiers and connectives has a very big impact on the ultimate meaning of a logical statement. For example, if X is an object in the universe of discourse (i.e., the set of objects being considered), the predicate $zebra(X)$ means X is a zebra and the predicate $striped(X)$ means X is striped, the statement

$$\forall x \text{ zebra}(x) \Rightarrow \text{striped}(x),$$

is read as *all zebras are striped*. Intuitively, you can think of logical implication as an *if-then* conditional statement, in which case this is read as *for all x in the universe of discourse, if x is a zebra, then x is striped*.

The meaning of this statement can be changed by changing the connective. For example, the statement

$$\forall x \text{ zebra}(x) \wedge \text{striped}(x),$$

is read as *everything is a zebra and everything is striped*, meaning everything in the universe of discourse is a striped zebra!

Similarly, changing the quantifier will also change the meaning. For example, the statement

$$\exists x \text{ zebra}(x) \wedge \text{striped}(x),$$

is read as *there exists an x , where x is a zebra and x is striped*, meaning that out of all of the objects in the universe of discourse *at least one* is a striped zebra. Finally, the statement

$$\exists x \text{ zebra}(x) \Rightarrow \text{striped}(x),$$

is a bit more tricky and you will almost never see this combination of quantifier and connective. This statement is read as *there exists an x , where if x is a zebra, then x is striped*. The difference is a bit subtle, but the important thing to understand is that this statement only guarantees that x exists, it does not guarantee that x is a zebra or that it is striped!

As a general rule-of-thumb, the universal quantifier ($\forall x$) is usually paired with the implication connective ($\Rightarrow$) and the existential quantifier ($\exists x$) is usually paired with the conjunction connective ($\wedge$).

PART 1: Predicate logic formulas

- (a) Choose a vocabulary of predicates, constants and functions appropriate for representing the following information in First Order Logic, then represent each sentence in FOL.
 - i. "There is someone who is loved by everybody"
 - ii. "All cats are lazy"
 - iii. "Some students are clever"
 - iv. "No student is rich"
 - v. "Every man loves a woman" (represent both meanings)
 - vi. "Everything is bitter or sweet"
 - vii. "Everything is bitter or everything is sweet"
 - viii. "Martin has a new bicycle"
 - ix. "Lynn gets a present from John, but she doesn't get any present from Peter"
- (b) Do the same as for the previous question.
 - i. Anyone who is rich is powerful.
 - ii. Anyone who is powerful is corrupt.
 - iii. Anyone who is meek and has a corrupt boss is unhappy.
 - iv. Not all students take both Computing-Theory and OO-Programming.
 - v. Only one student failed Computing-Theory.

- vi. Only one student failed both Computing-Theory and OO-Programming.
 - vii. The best score in Computing-Theory was better than the best score in OO-Programming.
 - viii. Every person who dislikes all donkeys is smart.
 - ix. There is a woman who likes all men who are not footballers.
 - x. There is a barber who shaves all men who do not shave themselves.
 - xi. No person likes a lecturer unless the lecturer is smart.
 - xii. Politicians can fool some of the people all of the time, and they can fool all of the people some of the time, but they can't fool all of the people all of the time.
- (c) Consider the following story:

I married a widow (let's call her w) who has a grown up daughter (d). My father (f), who visited us quite often, fell in love with my step-daughter and married her. Hence my father became my son-in-law and my step-daughter became my mother. Some months later, my wife gave birth to a son (s_1), who became the brother-in-law of my father, as well as my uncle. The wife of my father, that is, my step-daughter, also had a son (s_2).

Using predicate calculus, create a set of expressions that represent the situation in the above story. Extend the kinship relations to in-laws and uncle. Use generalised modus ponens to prove 'I am my own grandfather'.

PART 2: Representing family relationships.....

You are given an enormous table of family information. This consists of a very large number of facts, which are organised into predicates as below.

Predicate	Meaning
$parent(x, y)$	x is a parent of y
$male(x)$	x is male
$female(x)$	x is female

Use this information to define the relationships below by giving an appropriate formula based on the above three predicates.

Example: To capture the relation " x is the brother of y ", we can write:

$$male(x) \wedge \exists z (parent(z, x) \wedge parent(z, y))$$

and so define the new relation $brother(x, y)$ via the following formula (note the formula is an equivalence between the new predicate and the formula):

$$\forall x, y [brother(x, y) \Leftrightarrow (male(x) \wedge \exists z (parent(z, x) \wedge parent(z, y)))]$$

This formula is, basically, defining what the brother relation is in terms of the other two relations. Note the universal quantification at the outset of the whole equivalence formula: the relation between x and y holds for every pair of x and y .

- (a) x is the sister of y .
- (b) x is the uncle of y .
- (c) x is the grandmother of y .
- (d) x is an orphan.
- (e) x is not an orphan.
- (f) x is someone's mother.

- (g) x is someone's son.
- (h) x is a first cousin of y , that is, x and y have a common grandparent.
- (i) x is a first or second cousin of y .
- (j) x is an ancestor of y .
- (k) x is a descendant of y .
- (l) All of x 's grandchildren are male.
- (m) x does not have any grandparents.

If you are keen, you can try executing some similar definitions using the language *Prolog*. Download and install a Prolog interpreter (a good one is [SWI-Prolog](#)), and load the file *family.pl* available [here](#). The syntax may take a little getting used to, but in this file you will find various definitions like those in the previous question.

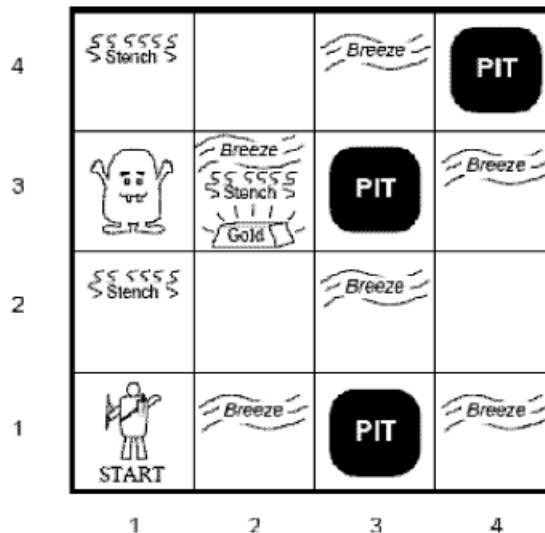
PART 3: Unification.....

- (a) Give the most general unifier (MGU) for the following pairs of expressions, or say why it does not exist. In all of these expressions variables are lower-case x , y , or z .
 - i. $p(a, b, b), p(x, y, z)$.
 - ii. $q(y, g(a, b)), q(g(x, x), y)$.
 - iii. $older(father(y), y), older(father(x), john)$.
 - iv. $knows(father(x), y), knows(x, x)$.
 - v. $r(f(a), b, g(f(a))), r(x, y, z)$.
 - vi. $older(father(y), y), older(father(y), john)$.
 - vii. $f(g(a, h(b)), g(x, y)), f(g(z, y), g(y, y))$.
- (b) Some more (y, w, z, v, u are all variables):
 - i. $p(f(y), w, g(z)), p(v, u, v)$.
 - ii. $p(f(y), w, g(z)), p(v, u, x)$.
 - iii. $p(a, x, f(g(y))), p(z, h(w), f(w))$.
 - iv. $p(z, h(w), g(z)), p(v, u, v)$.
 - v. $p(a, h(w), f(g(y))), p(z, x, f(w))$.

PART 4: Representation and reasoning.....

- (a) Write down logical representations for the following sentences, suitable for use with generalised modus ponens. Pay special attention to implications and quantification.
 - i. Horses, cows and pigs are mammals.
 - ii. An offspring of a horse is a horse.
 - iii. Bluebeard is a horse.
 - iv. Bluebeard is Charlie's parent.
 - v. Offspring and parent are inverse relations.
 - vi. Every mammal has a parent.
- (b) Using the logical expressions from the previous question do the following:
 - i. Draw the proof tree using backward chaining (that is, starting from your query) for the query "there exists a horse" - $(\exists x)Horse(x)$.
 - ii. Repeat the proof using forward chaining (that is, start with the facts and derive further conclusions). Do you get the the same results as before?
 - iii. How many solutions for H are there?

- (c) A popular children's riddle is “*Brothers and sisters I have none, but that man's father is my father's son*”. Use the rules of the kinship domain to work out who that man is.
- (d) For the following Wumpus world:



- i. Develop a notation capturing the important aspects of this domain in first order logic.
- ii. How would you express in a first order logic sentence:
 - α) If a square has no smell then the Wumpus is not in this square or any of the adjacent squares?
 - β) If there is stench in a square there must be a Wumpus in this square or in one of the adjacent squares?
- iii. Suppose the agent has traversed the path $(1, 1) \Rightarrow (1, 2) \Rightarrow (2, 2) \Rightarrow (1, 2)$. How can the agent deduce that the Wumpus is in square [1,3] using the laws of inference of first order logic?

PART 5: Resolution proofs

As with propositional logic, applying resolution to first-order logic formulas requires that they be first converted to conjunctive normal form. This is a bit more complicated when dealing with predicates however, because of the existential and universal quantifiers. We need to find a certain kind of reduction from first-order logic to propositional logic, in order to use the principles of resolution to prove our statements, and Herbrand's theorem provides us with such a reduction. However, this theorem applies to the domain closure of a set of formulas with no quantifier, so in order to apply it, we need a way to remove quantifiers from first-order logic formulas.

Basically, a first-order clause (disjunction of literals) is always assumed to be universally quantified. So, if a clause like $(P(x) \vee \neg Q(x, y) \vee R(y))$ stands for the sentence $\forall x, y. (P(x) \vee \neg Q(x, y) \vee R(y))$. So, when the only quantifiers we have are universal, this is easy: if we have a sentence $\forall x. P(x) \vee Q(y)$, its clausal representation is just $(P(x) \vee Q(y))$. Similarly, $\forall x. P(x) \Rightarrow Q(x)$ can be written in CNF as $(\neg P(x) \vee Q(x))$.

Where it becomes more complicated is when we are using the existential quantifier. If we have a formula $\exists x. P(x)$, we cannot just write clause $P(x)$, so that clause stands for $\forall x. P(x)$. This is where the idea of *Skolemisation* comes in. As $\exists x. P(x) \vee Q(x)$ says that “there exists some value of x ” for which $P(x) \vee Q(x)$ is true, we can assume a “witness” value representing such value, and hence use a constant A , so-called the *Skolem constant* for such x , for this value and write $P(A) \vee Q(A)$.

Now, what happens with a formula like $\forall x, \exists y. (P(x) \wedge Q(y))$? The fact is that the value of y that makes the formula true, will depend on each specific x . So, when dealing with formulas that mix quantifiers, we need to use a so-called *Skolem function*, to indicate that the element we are claiming to exist depend on the value of the previously universally quantified variables. A Skolem function is an n -place function, where n is the number of universal quantifiers that appear before the existentially quantified variable.

For example:

$$\begin{aligned} [\forall x.P(x) \Rightarrow \exists y.Q(y) \wedge R(x, y)] &\equiv [P(x) \Rightarrow Q(f(x)) \wedge R(x, f(x))] \\ [\forall x \forall y.P(x, y) \Rightarrow \exists z.Q(z) \wedge R(x, z)] &\equiv [P(x, y) \Rightarrow Q(f(x, y)) \wedge R(x, f(x, y))] \\ [\forall x.P(x) \Rightarrow \exists y \forall z.(Q(y) \wedge R(y, z))] &\equiv [P(x) \Rightarrow (Q(f(x)) \wedge R(f(x), z))] \\ [\exists x \forall y \forall z.P(x) \wedge (Q(y) \Rightarrow R(x, z))] &\equiv [P(A) \wedge (Q(y) \Rightarrow R(A, z))] \end{aligned}$$

In this last example, we use the Skolem constant A , as there is no universal quantifier that occurs before the existential quantifier. Of course, these examples are only used to demonstrate how to remove the quantifiers from these formulas; in order to apply resolution, we must also convert them into CNF.

When doing KRR below, we take information given in English and represent such information in first-order logic. We then suitably convert the resulting formulas so we can perform resolution, by transforming the formulas to CNF + Skolemization. Finally, we construct a resolution proof for the query of concern. There are fully work examples in the book (Section 9.5 - Resolution).

(a) Consider the following sentences:

1. Marcus was a man.
2. Marcus was a Roman.
3. All men are people.
4. Caesar was a ruler.
5. All Romans were either loyal to Caesar or hated him (or both).
6. Everyone is loyal to someone.
7. People only try to assassinate rulers they are not loyal to.
8. Marcus tried to assassinate Caesar.

Carry out the following tasks:

- i. Convert each of these sentences into first-order logic and then convert each formula into clausal form. Indicate any Skolem functions or constants used in the conversion.
- ii. Convert the negation of the question “Who hated Caesar?” into causal form (with an answer literal)
- iii. Derive the answer to this question using first-order resolution. Give the answer in English. In the proof use the notation developed in lectures.

(b) Determine whether the following sentence is valid (i.e., a tautology) using FOL Resolution:

$$\exists x \forall y \forall z ((P(y) \Rightarrow Q(z)) \Rightarrow (P(x) \Rightarrow Q(x))).$$

(c) (From [Brachman and Levesque 2005]) Victor has been murdered, and Arthur, Bertram, and Carleton are the only suspects (meaning exactly one of them is the murderer). Arthur says that Bertram was the victim’s friend, but that Carleton hated the victim. Bertram says that he was out of town the day of the murder, and besides, he didn’t even know the guy. Carleton says that he saw Arthur and Bertram with the victim just before the murder. You may assume that everyone—except possibly for the murderer—is telling the truth.

You are also given the following general facts:

- anyone who was with Victor on the day of the murder, must have been in town;
- a person who is with someone, must also know them;
- a friend knows the person who they call a friend;
- a person who hates someone must also know them.

You should use the following set of predicates when expressing the general facts and statements made by the suspects:

- $I(x)$: means x is innocent;
- $F(x, y)$: means x is a friend of y ;

- $H(x, y)$: means x hates y ;
 - $T(x)$: means x is in town;
 - $K(x, y)$: means x knows y ; and
 - $W(x, y)$: means x is with y .
- i. Use Resolution to find the murderer. In other words, formalize the facts as a set of clauses, prove that there is a murderer (here, we assume that there is only one murderer), and extract their identity from the derivation.
 - ii. Suppose we discover that we were wrong—we cannot assume that there was only a single murderer (there may have been a conspiracy). Show that in this case the facts do not support anyone's guilt. In other words, for each suspect, present a logical interpretation that supports all the facts, but where that suspect is innocent (and maybe the other two are guilty!).

End of tutorial worksheet